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APPARATUS AND METHOD FOR UPDATING SOFTWARE IN A PATIENT SUPPORT APPARATUS USING A MEMORY TOGGLE

Abstract

An apparatus, system or method includes a patient bed for use with a network including a remote computer. The patient bed includes a bed frame to support a patient and bed circuitry carried by the bed frame. The bed circuitry is configured to receive a new version of bed operating software from the remote computer. The bed circuitry includes a first memory bank and a second memory bank. One of the first and second memory banks stores therein a current version of bed operating software. The other of the first and second memory banks stores therein the new version of bed operating software received from the remote computer. Thus, two versions of bed operating software are stored in the bed circuitry. A memory toggle is used to determine which version of the operating software in which of the two memory banks is used to operate the bed.

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Background/Summary

[0001] The present application is a continuation of U.S. application Ser. No. 16/502,111, filed Jul. 3, 2019, now U.S. Patent No. XXXXXXXX, which claims the benefit, under 35 U.S.C. § 119(e), of U.S. Provisional Application No. 62/711,102, filed Jul. 27, 2018, each of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] The present disclosure relates to patient support apparatuses such as patient beds and particularly, to an apparatus, system and method for updating software of a patient support apparatus. More particularly, the present disclosure relates to transmitting updated software from a remote computer to multiple patient support apparatuses.

[0003] Patient support apparatuses such as patient beds found in healthcare facilities are sometimes upgraded with new operating software. The operating software is sometimes referred to as firmware. To upgrade the software of some patient support apparatuses, a service technician walks from bed-to-bed and physically connects a computer, such as a laptop or tablet computer, to the bed for replacement of the bed's current operating software with the new operating software. Some patient support apparatuses are able to have the operating software upgraded from a remote computer. See, for example, U.S. Pat. No. 9,513,899 in this regard. However, in the prior art, the patient support apparatuses are rendered inoperable during the software upgrade process while the existing or current version of the operating software is being overwritten with the new version of the operating software.

[0004] In some prior art systems, the software upgrade process can take up to twenty minutes to complete. During that time, any patients assigned to the beds being upgraded may need to be taken out of the respective beds, particularly if the bed has an air mattress which may deflate while the bed is inoperable. Furthermore, when the patient support apparatuses are being upgraded with new operating software by a remote computer, the caregiver at the remote computer needs to coordinate with other caregivers to identify which beds in which rooms are getting ready to have the software upgraded so the patients can be removed from the respective beds at the appropriate time. Accordingly, there is a need to improve the manner in which patient support apparatuses are upgraded with new versions of operating software.

SUMMARY

[0005] The present application discloses one or more of the features recited in the appended claims and/or the following features which, alone or in any combination, may comprise patentable subject matter:

[0006] A patient bed for use with a network including a remote computer may be provided and may include a bed frame to support a patient and bed circuitry that may be carried by the bed frame. The bed circuitry may be configured to receive a new version of bed operating software from the remote computer. The bed circuitry may include a first memory bank and a second memory bank. One of the first and second memory banks may store therein a current version of bed operating software. The other of the first and second memory banks may store therein the new version of bed operating software that may be received from the remote computer. Thus, two versions of bed operating software may be stored in the bed circuitry.

[0007] In some embodiments, the bed circuitry may be programmed with a memory toggle that may be used to select between the first and second memory banks thereby to select which version of bed operating software is used to operate the patient bed. The first and second memory banks each may include flash memory banks that are independent of each other. Alternatively or additionally, the memory toggle may include a toggle flag that may be used at start-up of the patient bed to determine which of the first and second memory banks is currently active.
[0008] It is contemplated by this disclosure that the new version of bed operating software received from the remote computer may be stored at download into whichever of the first and second memory banks may be inactive. Upon next start-up of the patient bed, the memory toggle may be switched so that the previously active memory bank becomes inactive and so that the previously inactive memory bank becomes active. If desired, during download of the new version of bed operating software into the inactive memory bank, the patient bed may continue to be operated according to the current version of bed operating software stored in the active memory bank.
[0009] Optionally, after the patient bed is operated according to the new version of the bed operating software, the bed circuitry may be programmed to automatically switch the memory toggle if an error occurs in the new version of bed operating software so that the prior version of bed operating software may once again be used to operate the patient bed. Alternatively or additionally, a user input may be carried by the frame and the user input may be usable to manually switch the memory toggle so that a user is able to select which of the software versions stored in the first and second memory banks is used to operate the patient bed. Further alternatively or additionally, the remote computer may be used to command the bed circuitry to switch the memory toggle to change which software version stored in the first and second memory banks is used to operate the patient bed.
[0010] According to another aspect of the present disclosure, a system may include a computer and a plurality of patient beds that may be remote from the computer. Each patient bed may include a bed frame to support a patient and bed circuitry that may be carried by the bed frame. The bed circuitry of each bed may be configured to receive a new version of bed operating software from the remote computer. The bed circuitry may include a first memory bank and a second memory bank. One of the first and second memory banks may store therein a current version of bed operating software for each bed. The other of the first and second memory banks may store therein the new version of bed operating software that may be received from the remote computer. Thus, two versions of bed operating software may be stored in the bed circuitry of each bed. If desired, the computer may be configured to upgrade the plurality of patient beds with the new version of bed operating software simultaneously.
[0011] In some embodiments, the computer may be configured to display each of the plurality of patient beds that may be available for upgrade. Alternatively or additionally, each of the patient beds that may be available for upgrade may register with the computer. Each of the patient beds that register with the computer may transmit a bed identification (ID) and a software version number of the current bed operating software for the respective patient bed.
[0012] Optionally, the computer may be configured to permit a user to select which of the plurality of patient beds is to be upgraded with the new version of bed operating software and to deselect others of the patient beds that are not to be upgraded. Further optionally, the bed circuitry of each patient bed may be programmed with a memory toggle that may be used to select between the first and second memory banks thereby to select which version of bed operating software is used to operate the respective patient bed. The first and second memory banks of each patient bed each may include flash memory banks that may be independent of each other.
[0013] In some embodiments, the memory toggle of each patient bed may include a toggle flag that mat be used at start-up of the respective patient bed to determine which of the first and second memory banks is currently active. The new version of bed operating software received from the remote computer by each patient bed may be stored at download into whichever of the first and

second memory banks is inactive in the respective patient bed. Upon next start-up of each of the patient beds, the memory toggle may be switched so that the previously active memory bank becomes inactive and so that the previously inactive memory bank becomes active. During download of the new version of bed operating software into the inactive memory bank of each patient bed, each of the patient beds may continue to be operated according to the current version of bed operating software stored in the active memory bank of the respective patient bed.

[0014] If desired, after each of the patient beds is operated according to the new version of the bed operating software, the respective bed circuitry may be programmed to automatically switch the memory toggle if an error occurs in the new version of bed operating software so that the prior version of bed operating software is once again used to operate the respective patient bed. Alternatively or additionally, a user input may be carried by the frame of each patient bed. The respective user inputs may be usable to manually switch the memory toggle of the respective patient bed so that a user is able to select which of the software versions stored in the first and second memory banks of the respective bed may be used to operate the respective patient bed. Further alternatively or additionally, the remote computer may be used to command the bed circuitry to switch the memory toggle to change which software version stored in the first and second memory banks is used to operate the patient bed.

[0015] According to a further aspect of the present disclosure, a method may include providing a patient bed that may have a bed frame to support a patient, providing bed circuitry that may be carried by the bed frame, storing current bed operating software in a first memory bank of the bed circuitry, and transmitting a new version of bed operating software from a remote computer to the bed circuitry for storage in a second memory bank of the bed circuitry. Thus, two versions of bed operating software may be stored in the bed circuitry according to the method.

[0016] In some embodiments, the method may further include providing the bed circuitry with a memory toggle that may be used to select between the first and second memory banks thereby to select which version of bed operating software may be used to operate the patient bed. The first and second memory banks each may comprise flash memory banks that are independent of each other, for example. Alternatively or additionally, the memory toggle may include a toggle flag that may be used at start-up of the patient bed to determine which of the first and second memory banks is currently active. Further alternatively or additionally, the method includes using a memory toggle at the remote computer to command the bed circuitry to switch between the first and second memory banks thereby to select which version of bed operating software is used to operate the respective patient bed.

[0017] In some instances, the second memory bank may be inactive when the new version of bed operating software is received from the remote computer. The method may further include switching the memory toggle upon next start-up of the patient bed so that the previously active memory bank becomes inactive and so that the previously inactive memory bank becomes active. The method may also include continuing to operate the patient bed according to the current version of bed operating software during download of the new version of bed operating software into the inactive memory bank.

[0018] Optionally, the method may further include, after the patient bed is operated according to the new version of the bed operating software, automatically switching the memory toggle if an error occurs in the new version of bed operating software so that the prior version of bed operating software may once again be used to operate the patient bed. Alternatively or additionally, the method may further include using a user input carried by the frame to manually switch the memory toggle to select which of the software versions stored in the first and second memory banks may be used to operate the patient bed.

[0019] According to still another aspect of the present disclosure, a method may include providing a computer and providing a plurality of patient beds that may be remote from the computer. Each patient bed may include a bed frame to support a patient and bed circuitry that may be carried by the bed frame. The method may further include storing current bed operating software in a first memory bank of the bed circuitry of each of the patient beds and transmitting a new version of bed operating software from the remote computer to the bed circuitry for storage in a second memory bank of the bed circuitry of each of the patient beds. Thus, two versions of bed operating software may be stored in the bed circuitry of each of the patient beds.

[0020] In some embodiments, the method may further include displaying each of the plurality of patient beds that may be available for upgrade on a display screen of the remote computer. If desired, the method may further include registering each of the patient beds that may be available for upgrade with the computer. The method may also include transmitting to the remote computer a bed identification (ID) and a software version number of the current bed operating software from each of the patient beds that may register with the remote computer. Optionally, the method may further include selecting with the remote computer which of the plurality of patient beds may be upgraded with the new version of bed operating software and deselecting with the computer others of the patient beds that are not to be upgraded.

[0021] It is contemplated by this disclosure that the method may further include using a memory toggle of each patient bed to select between the first and second memory banks thereby to select which version of bed operating software is used to operate the respective patient bed. The first and second memory banks of each patient bed each may include flash memory banks that are independent of each other. Alternatively or additionally, the memory toggle of each patient bed may include a toggle flag that may be used at start-up of the respective patient bed to determine which of the first and second memory banks may be currently active.

[0022] In some embodiments, the new version of bed operating software that may be received from the remote computer by each patient bed may be stored at download into whichever of the first and second memory banks may be inactive in the respective patient bed. The method may further include upon next start-up of each of the patient beds, automatically switching the memory toggle so that the previously active memory bank may become inactive and so that the previously inactive memory bank may become active for each of the patient beds. Optionally, the method may further include continuing to operate the patient bed according to the current version of bed operating software during download of the new version of bed operating software into the inactive memory bank.

[0023] Optionally, the method may further include, after each patient bed is operated according to the new version of the bed operating software, automatically switching the memory toggle if an error occurs in the new version of bed operating software so that the prior version of bed operating software may once again be used to operate the respective patient bed. Alternatively or additionally, the method may further include using a user input that may be carried by the frame to manually switch the memory toggle to select which of the software versions stored in the first and second memory banks may used to operate the patient bed.

[0024] Additional features, which alone or in combination with any other feature(s), such as those listed above and/or those listed in the claims, may comprise patentable subject matter and will become apparent to those skilled in the art upon consideration of the following detailed description of various embodiments exemplifying the best mode of carrying out the embodiments as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The detailed description particularly refers to the accompanying figures, in which:

[0026] FIG. **1** is a diagrammatic view of a bed operating software upgrade system showing a patient bed coupled to a network via a bed connector; the bed being in communication with an upgrade server, a communication server, and a real time locating system (RTLS) server via the network; and a receiver of the RTLS being coupled to the RTLS server and in communication with a locating badge carried by a caregiver that is located alongside the patient bed;

[0027] FIG. **2** is a block diagram showing various components of the patient bed and the bed operating software upgrade system;

[0028] FIG. **3** is a diagrammatic view showing a plurality of the patient beds coupled to the network and showing one of the beds having a first memory bank that is active and a second memory bank that is inactive as determined by a first state of a memory toggle prior to a bed operating software upgrade and showing the memory toggle in a second state after the bed operating software upgrade so that the first memory bank is inactive and the second memory bank is active;

[0029] FIG. **4** is a screen shot of an upgrade computer screen showing a list of beds that are available for a software upgrade, showing a software version of the current bed operating software, and showing a set of selection icons or buttons for selecting which of the patient beds are to bed upgraded with new bed operating software; and

[0030] FIG. **5** is a screen shot of a remote toggle screen showing a list of beds that have experienced operating software malfunction errors, showing the software versions that are stored in the active and inactive memory banks for each of the listed beds, and showing a set of selection icons or buttons for selecting whether to toggle the memory banks so that the inactive memory bank becomes the active memory bank for the respective bed resulting in a different bed operating software version becoming active for the respective bed.

DETAILED DESCRIPTION

[0031] A bed operating software upgrade system **10** includes a plurality of patient beds **12**, only one of which is shown in FIG. **1**, coupled to a network **18** of a healthcare facility such as a hospital, outpatient care facility, nursing home, and the like. In the illustrative example, patient bed **12** is coupled to network **18** via a bed connector unit **14** in the respective patient room. Illustratively, bed **12** is coupled to unit **14** with a cable **16** such as a cable having 37-pin connectors at its opposite ends as is known in the art. In other embodiments, a wireless connection is made between bed **12** and unit **14**. Examples of suitable units **14** for use with bed **12** include bed interface units (BIU's), network interface unit (NIU's), and wireless interface unit (WIU's) available from Hill-Rom Company, Inc. of Batesville, Indiana.

[0032] Further details of BIU's and NIU's are shown and described in U.S. Pat. Nos. 7,538,659 and 7,319,386 and in U.S. Patent Application Publication Nos. 2009/0217080 A1, 2009/0212925 A1 and 2009/0212926 A1, each of which is hereby expressly incorporated by reference herein to the extent not inconsistent with this disclosure which shall control as to any inconsistencies. Further details of WIU's are shown and described in U.S. Patent Application Publication No. 2007/0210917 A1 which is also hereby expressly incorporated by reference herein to the extent not inconsistent with this disclosure which shall control as to any inconsistencies. Other types of bed connector units **14** are also within the scope of this disclosure including, but not limited to, a communications hub (aka “Universal Collector”) shown and described in U.S. Patent Application Publication No. 2017/0323555 A1 and which is hereby incorporated herein by reference herein to the extent not inconsistent with this disclosure which shall control as to any inconsistencies. Therefore, the bed connector unit **14** shown in FIG. **1** is intended to represent all types of connectors that are used to receive bed status data from bed **12** and to communicate data from one or more other computer devices to bed **12** within a network **18**.

[0033] As shown diagrammatically in FIGS. **1** and **2**, a communication server **20** of a corresponding communication system **21**, an upgrade server **22** of a corresponding upgrade system **23**, and a real time locating system (RLTS) server **24** of a corresponding RTLS **25** are each coupled to network **18**. Double headed arrows **26** in FIG. **1** represent the bidirectional communication links between network **18**, unit **14** (and the corresponding bed **12**) and each of servers **20**, **22**, **24** and therefore, with each other. Communication links **26** include wired communication links or wireless communication links at the option of the designer of system **10** in any given healthcare facility. Thus, in some embodiments, bed connector **14** includes wireless communication circuitry **15**. Circuitry **15** communicates wirelessly with patient bed **12** and/or with network **18** such as through a wireless access point (WAP) (not shown) of network **18**. Such wireless communication contemplated by this disclosure includes Bluetooth (BT), Bluetooth Low Energy (BLE), Zigbee, Z-Wave, and WiFi (e.g., any of the 802.11x protocols). However, it should be understood that all types of wireless communication are within the scope of the present disclosure, including infrared (IR) communications, ultrasonic (US) communications, ultra-wideband (UWB) communications, and so forth. In some embodiments in which bed **12** communicates wirelessly with bed connector **14**, cable **16** is omitted.

[0034] The RTLS **25** of system **10** includes wireless transceiver units **28** placed throughout the healthcare facility. Only one such unit **28** is depicted diagrammatically in FIG. **1**. The RTLS **25** of system **10** also includes caregiver locating or tracking tags or badges **30** that are worn by caregivers. Each of the transceiver units **28** receives a wireless signal from the badges **30** of each of the caregivers wearing badges **30** and that are within communication range of the respective unit **28** as indicated diagrammatically by arrow **32** in FIG. **1**. The wireless signal from each badge **30** includes badge identification data (ID) which is unique to the corresponding badge **30**. Unit **28** then transmits its ID, which corresponds to a particular location in the healthcare facility, and the badge (ID) to RTLS server **24** as indicated diagrammatically by a bidirectional communication link **34** in FIG. **1**. Based on the received badge ID and the location ID from unit **28**, server **24** determines the location of the caregiver within the healthcare facility. The location of assets such as beds **12** is tracked in a similar manner by server **24**, in some embodiments, by attaching asset tags that are substantially the same as badges **30** to the assets to be tracked. In FIG. **1**, the caregiver is shown carrying a tablet computer **38** which is configured to communicate wirelessly with other devices of network **18**.

[0035] The present disclosure is primarily focused on updating or upgrading beds **12** with new operating software from server **22**. The terms “upgrade” and “update” and other forms thereof (e.g., “upgrading” and “updating”) are used interchangeably herein. Upgraded or updated operating software is intended to refer to new software for the patient support apparatus **12** to which it is sent, without regard to whether the new operating software is better than the prior version of the operating software in any respect. Server **22** is configured to upgrade the software of multiple beds **12** simultaneously as will be discussed in further detail below. To provide context to the types of features and functions to which bed operating software pertains, a discussion is provided below of the basic components and operation of various features of bed **12**. It should be understood, however, that the particulars of any given version of software (e.g., the actual software code, itself) is not needed in order to understand the present disclosure. That is, bed operating software is the software that is executed to operate various electronic features and functions of bed **12**.

[0036] Bed **12** includes a patient support structure such as a frame **40** that supports a surface or mattress **42** as shown in FIG. **1**. It should be understood that FIG. **1** shows some details of one possible bed **12** having one particular configuration. However, this disclosure is applicable to other types of patient support apparatuses **12** having other configurations, including other types of beds, surgical tables, examination tables, stretchers, chairs, wheelchairs, patient lifts and the like.

[0037] Still referring to FIG. **1**, frame **40** of bed **12** includes a base frame **48** (sometimes just referred herein to as a base **48**), an upper frame assembly **50** and a lift system **52** coupling upper frame assembly **50** to base **48**. Lift system **52** is operable to raise, lower, and tilt upper frame assembly **50** relative to base **48**. Bed **12** has a head end **54** and a foot end **56**. Patient bed **12** further includes a footboard **55** at the foot end **56** and a headboard **57** at the head end **54**. Illustrative bed **12** includes a pair of push handles **47** coupled to an upstanding portion **59** of base **48** at the head end **54** of bed **12**. Only a portion of one push handle **47** can be seen in FIG. **1**. Headboard **57** is coupled to upstanding portion **59** of base as well. Footboard **55** is coupled to upper frame assembly **50**. Base **48** includes wheels or casters **49** that roll along floor (not shown) as bed **12** is moved from one location to another. A set of foot pedals **51** are coupled to base **48** and are used to brake and release casters **49**.

[0038] Illustrative patient bed **12** has four siderail assemblies coupled to upper frame assembly **50** as shown in FIG. **1**. The four siderail assemblies include a pair of head siderail assemblies **58** (sometimes referred to as head rails) and a pair of foot siderail assemblies **60** (sometimes referred to as foot rails). Each of the siderail assemblies **58** and **60** is movable between a raised position, as shown in FIG. **1**, and a lowered position (not shown). Siderail assemblies **58**, **60** are sometimes referred to herein as siderails **58**, **60**. Each siderail **58**, **60** includes a barrier panel **64** and a linkage **66**. Each linkage **66** is coupled to the upper frame assembly **50** and is configured to guide the barrier panel **64** during movement of siderails **58**, **60** between the respective raised and lowered positions. Barrier panel **64** is maintained by the linkage **66** in a substantially vertical orientation during movement of siderails **58**, **60** between the respective raised and lowered positions.

[0039] Upper frame assembly **50** includes various frame elements **68** that form, for example, a lift frame and a weigh frame supported with respect to the lift frame by a set of load cells **72** of a scale and/or patient position monitoring (PPM) system **70** of bed **12**. A patient support deck **74**, shown diagrammatically in FIG. **2**, is carried by the weigh frame portion of upper frame assembly **50** and supports mattress **42** thereon. Operation of the scale/PPM system **70** is among the features of bed **12** controlled by the bed operating software.

[0040] Patient support deck **74** includes a head section **80**, a seat section **82**, a thigh section **83** and a foot section **84** in the illustrative example as shown diagrammatically in FIG. **2**. Sections **80**, **83**, **84** are each movable relative to the weigh frame portion of upper frame assembly **50**. For example, head section **80** pivotably raises and lowers relative to seat section **82** whereas foot section **84** pivotably raises and lowers relative to thigh section **83**. Additionally, thigh section **83** articulates relative to seat section **82**. Also, in some embodiments, foot section **84** is extendable and retractable to change the overall length of foot section **84** and therefore, to change the overall length of deck **74**. For example, foot section **84** includes a main portion **85** and an extension **87** in some embodiments as shown diagrammatically in FIG. **2**.

[0041] In the illustrative embodiment, seat section **82** is fixed in position with respect to the weigh frame portion of upper frame assembly **50** as patient support deck **74** moves between its various patient supporting positions including a horizontal position to support the patient in a supine position, for example, and a chair position (not shown) to support the patient in a sitting up position. In other embodiments, seat section **82** also moves relative to upper frame assembly **50**, such as by pivoting and/or translating. Of course, in those embodiments in which seat section **82** translates relative to the upper frame assembly **50**, the thigh and foot sections **83**, **84** also translate along with seat section **82**. As bed **12** moves from the horizontal position to the chair position, foot section **84** lowers relative to thigh section **83** and shortens in length due to retraction of the extension **87** relative to main portion **85**. As bed **12** moves from the chair position to the horizontal position, foot section **84** raises relative to thigh section **83** and increases in length due to extension of the extension **87** relative to main portion **85**. Thus, in the chair position, head section **80** extends upwardly from upper frame assembly **50** and foot section **84** extends downwardly from thigh section **83**.

[0042] As shown diagrammatically in FIG. **2**, bed **12** includes a head motor or actuator **90** coupled to head section **80**, a knee motor or actuator **92** coupled to thigh section **83**, a foot motor or actuator **94** coupled to foot section **84**, and a foot extension motor or actuator **96** coupled to foot extension **87**. Motors **90**, **92**, **94**, **96** may include, for example, an electric motor of a linear actuator. In those embodiments in which seat section **82** translates along upper frame assembly **50** as mentioned above, a seat motor or actuator (not shown) is also provided. Head motor **90** is operable to raise and lower head section **80**, knee motor **92** is operable to articulate thigh section **83** relative to seat section **82**, foot motor **94** is operable to raise and lower foot section **84** relative to thigh section **83**, and foot extension motor **96** is operable to extend and retract extension **87** of foot section **84** relative to main portion **85** of foot section **84**. Operation of motors **90**, **92**, **94**, **96** is among the features of bed **12** controlled by the bed operating software.

[0043] In some embodiments, bed **12** includes a pneumatic system **98** that controls inflation and deflation of various air bladders or cells of mattress **42**. The pneumatic system **98** is represented in FIG. **2** as a single block but that block **98** is intended to represent one or more air sources (e.g., a fan, a blower, a compressor) and associated valves, manifolds, air passages, air lines or tubes, pressure sensors, and the like, as well as the associated electric circuitry, that are typically included in a pneumatic system for inflating and deflating air bladders of mattresses. Operation of pneumatic system **98** is among the features of bed **12** controlled by the bed operating software.

[0044] As also shown diagrammatically in FIG. **2**, lift system **52** of bed **10** includes one or more elevation system motors or actuators **100**, which in some embodiments, comprise linear actuators with electric motors. Thus, actuators **100** are sometimes referred to herein as motors **100** and operation of the motors **100** is among the features of bed **12** controlled by the bed operating software. Alternative actuators or motors contemplated by this disclosure include hydraulic cylinders and pneumatic cylinders, for example. The motors **100** of lift system **52** are operable to raise, lower, and tilt upper frame assembly **50** relative to base **48**. In the illustrative embodiment, one of motors **100** is coupled to, and acts upon, a set of head end lift arms **102** and another of motors **100** is coupled to, and acts upon, a set of foot end lift arms **104** to accomplish the raising, lowering and tilting functions of upper frame **50** relative to base **48**. Guide links **105** are coupled to base **48** and to lift arms **104** in the illustrative example as shown in FIG. **1**. Illustrative bed **12** of FIG. **1** is the CENTRELLA™ bed available from Hill-Rom Company, Inc. Other aspects of illustrative bed **12** are shown and described in more detail in U.S. Patent Application Publication No. 2018/0161225 A1 which is hereby expressly incorporated by reference herein to the extent not inconsistent with the present disclosure which shall control as to any inconsistencies.

[0045] Each of siderails **58** includes a first user control panel **106** coupled to the outward side of the associated barrier panel **64**. Controls panels **106** include various buttons that are used by a caregiver to control associated functions of bed **12**. For example, control panel **106** includes buttons that are used to operate head motor **90** to raise and lower the head section **80**, buttons that are used to operate knee motor **92** to raise and lower the thigh section **83**, and buttons that are used to operate motors **100** to raise, lower, and tilt upper frame assembly **50** relative to base **48**. In some embodiments, control panel **106** also includes buttons that are used to operate motor **94** to raise and lower foot section **84** and buttons that are used to operate motor **96** to extend and retract foot extension **87** relative to main portion **85**. Each of siderails **58** also includes a second user control panel **108** coupled to the inward side of the associated barrier panel **64**. Controls panels **108** include various buttons that are used by a patient to control associated functions of bed **12**. In some embodiments, the buttons of control panels **108**, **108** comprise membrane switches that are used to control head motor **90** and knee motor **92**.

[0046] As shown diagrammatically in FIG. **2**, bed **12** includes control circuitry **110** that is electrically coupled to motors **90**, **92**, **94**, **96** and to motors **100** of lift system **52**. Control circuitry **110** is represented diagrammatically as a single block in FIG. **2**, but control circuitry **110** in some embodiments, comprises various circuit boards, electronics modules, and the like that are electrically and communicatively interconnected. Control circuitry **110** includes one or more microprocessors **112** or microcontrollers that execute software to perform the various control functions and algorithms described herein. Thus, circuitry **110** also includes memory **114** for storing software, variables, calculated values, and the like as is well known in the art.

[0047] According to the present disclosure, memory **114** of circuitry **110** includes a first memory bank **111** and a second memory bank **113** which are dedicated for the storage of two versions of bed operating software. Thus, memory bank **111** stores a first version of bed operating software and memory bank **113** stores a second version of bed operating software. However, as will be discussed in further detail below, only of memory banks **111**, **113** is active at any given instance in time, with the other memory bank **113** being inactive. In some embodiments, memory banks **111**, **113** comprise flash memory banks that are independent of each other. That is, when memory bank **111** is active, none of the software in memory bank **113** will be used, and when memory bank **113** is active, none of the software in memory bank **111** will be used. In some embodiments, memory banks **111**, **113** are included in separate integrated circuit chips, such as EEPROM's, EPROM's, and the like. In other embodiments, memory banks **111**, **113** are included on the same integrated circuit chip but there is a dedicated block of memory addresses for each memory bank **111**, **113**.

[0048] As shown diagrammatically in FIG. **2**, a user inputs block represents the various user inputs such as buttons of control panels **106**, **108**, for example, that are used by the caregiver or patient to communicate input signals to control circuitry **110** of bed **12** to command the operation of the various motors **90**, **92**, **94**, **96**, **100** of bed **12**, as well as commanding the operation of other functions of bed **12**. Bed **12** includes at least one graphical user input (GUI) or display screen **120** coupled to a respective siderail **58** as shown in FIG. **1**. Display screen **120** is coupled to control circuitry **110** as shown diagrammatically in FIG. **2** and may be a touchscreen display, for example. In some embodiments, two graphical user interfaces **120** are provided and are coupled to respective siderails **58**. Alternatively or additionally, one or more graphical user interfaces are coupled to siderails **60** and/or to one or both of the headboard **57** and footboard **55**.

[0049] In the illustrative embodiment, bed **12** has a communication interface or port **116** which provides bidirectional communication via cable **16** with bed connector **14** which, in turn, communicates bidirectionally with network **18** and the various computers and systems of network **18**, such as illustrative servers or computers **21**, **23**, **25**. In other embodiments, communication interface **116** is used for wireless communications with bed connector **14**. Communication interface **116** also may be configured for wireless communication with network **18** and its associated devices without the use of bed connector **14** in some embodiments. For example, WiFi communications between communication interface **116** and one or more WAP's of network **18** is contemplated by this disclosure. The communication data to and from interface **116** is among the features of bed **12** controlled by the bed operating software.

[0050] Still referring to FIG. **2**, bed **12** includes various sensors to sense the states or positions of various portions of bed **12**. In the illustrative example, bed **12** includes an angle sensor **118** coupled to head section **80** to sense an angle of head section elevation (sometimes referred to as the head-of-bed (HOB) angle). Angle sensor **118** is an accelerometer (single-axis or multi-axis) in some embodiments. In such embodiments, the HOB angle is measured with respect to a horizontal reference axis and/or with respect to a vertical reference axis depending upon the orientation of the accelerometer relative to head section **80** and depending upon the type of accelerometer. In other embodiments, angle sensor **118** includes a rotary potentiometer which measures the HOB angle between head section **90** and another portion of frame **40** such as one of frame members **68** of upper frame assembly **50**. In further embodiments, angle sensor **90** is included in head motor **90** and has an output that correlates to the HOB angle. Motor **90** may include, for example, a shaft encoder, a Hall effect sensor, a rotary potentiometer, or some other sensor which serves as angle sensor **118** of bed **12** in such embodiments. Similar such sensors are included in elevation system motors **100** in some embodiments and are used to determine the position of upper frame assembly **50** relative to base **48** such as the height of upper frame assembly **50** and/or amount of tilt of upper frame assembly **50** relative to base **48**.

[0051] Bed **12** also includes siderail position sensors **122** to sense the position (e.g., raised and/or lowered) of each of siderails **58**, **60** and one or more caster braking sensors **124** to sense whether casters **49** are braked or released. In some embodiments, sensors **122**, **124** include limit switches that are engaged or disengaged by a linkage mechanism, such as linkage **66** in the case of siderails **58**, **60**, to produce output signals indicative of the position of the respective mechanical structure.

Alternatively, Hall effect sensors may be used as some or all of sensors **122**, **124** in some embodiments. The foregoing types of sensors **122**, **124** are just a couple examples of suitable sensors and therefore, this disclosure is intended to cover all types of sensors that may be used as sensors **122**, **124**. Each of the sensors mentioned above, including sensors internal to motors **100** and sensors **118**, **122**, **124** are each coupled electrical to control circuitry **110** for analysis and/or processing. Thus, data from sensors **118**, **122**, **124** is used by the bed operating software in connection with the control and operation of various features of bed **12**.

[0052] In some embodiments, bed **12** has safety protocol capability. Thus, control circuitry **110** is programmed with the bed operating software to enable and disable the safety protocols of bed **12**. It is within the scope of the present disclosure for the safety protocols to be enabled (aka “activated” or “turned on”) and disabled (aka “deactivated” or “turned off”) either remotely via a computer of network **18** or by use of user inputs **106**, **108**, **120** of bed **12**. In some embodiments, control circuitry **110** of bed **12** is configured to implement three different safety protocols, namely, a falls risk protocol (aka a falls protocol), a pulmonary protocol, and a safe skin protocol (aka a skin protocol). It should be understood that these are just examples of possible protocols for implementation on bed **12** and other protocols based on bed status information are within the scope of this disclosure. The safety protocols are among the features of bed **12** to which the bed operating software pertains.

[0053] As shown in FIG. **1**, bed **12** includes three status or alert lights at foot end **56** corresponding to the three protocols. Thus, bed **12** includes a falls alert light **126**, a pulmonary alert light **128**, and a skin alert light **130**. In the illustrative example, alert lights **126**, **128**, **130** are coupled to a lateral frame member of extension **87** of foot section **84** and are situated beneath footboard **55**. In other embodiments, alert lights **126**, **128**, **130** may be located elsewhere on bed **12** such as on base **48** and/or one or more of siderails **58**, **60**. In FIG. **2**, alert lights **126**, **128**, **130** are represented diagrammatically as a single block and are coupled electrically to control circuitry **110** to control the manner in which alert lights **126**, **128**, **130** are illuminated as will be discussed in further detail below. In other embodiments, bed **12** is equipped with the alert lights discussed in U.S. Patent Application Publication No. 2018/0161225 A1 which is already incorporated by reference herein, in lieu of or in addition to alert lights **126**, **128**, **130**.

[0054] In some embodiments, alert lights **126**, **128**, **130** are illuminated different colors to indicate certain statuses. For example, lights **126**, **128**, **130** are turned off if the particular protocol is not enabled, meaning the bed statuses contributing to the particular protocol are not being monitored for a protocol violation. Lights **126**, **128**, **130** are illuminated a first color, such as green for example, if the associated protocol is enabled, meaning the bed statuses contributing to the particular protocol are being monitored for a protocol violation, but all of the monitored bed statuses for the particular protocol are satisfactory or in a desirable state (i.e., not violated). Lights **126**, **128**, **130** are illuminated a second color, such as amber or yellow for example, if the associated protocol is enabled and at least one of the monitored bed statuses for the particular protocol is undesirable or unsatisfactory (i.e., violated).

[0055] In some embodiments, an audible alarm of bed **12** may also sound under the control of control circuitry **110** if an unsatisfactory condition of a particular protocol is detected. Lights **126**, **128**, **130** are illuminated a third color, such as blue for example, if the associated protocol is enabled and at least one of the monitored bed statuses for the particular protocol is undesirable (i.e., violated), but the alert has been suspended as will be discussed in further detail below. If the alert has been suspended, any associated audible alarms may be turned off during the alarm suspension. The operation of alert lights **126**, **128**, **130** and the audible alarm, if any, of bed **12** is among the features controlled by the bed operating software.

[0056] As mentioned above, illustrative bed **12** is configured to implement a falls risk protocol, a pulmonary protocol, and a safe skin protocol. According to the falls risk protocol example given herein, PPM system **70** is required to be enabled to monitor a position of the patient relative to the frame **40**, upper frame assembly **50** of frame **40** is required to be in a lowest position relative to a base **48** of frame **40** as sensed by sensors of motors **100**, siderails **58**, **60** of the frame **40** are required to be in the respective raised positions as sensed by siderail position sensors **122**, and the caster brake of one or more of casters **49** is required to be braked or set as sensed by caster braking sensors **124**. Thus, if PPM system **70** indicates that the patient is moving toward exiting bed **12** by a threshold amount (including exiting the bed **12**) such movement beyond the threshold is considered to be a violation of the falls protocol by control circuitry **110** assuming the falls protocol is enabled. Similarly, if upper frame assembly **50** is moved upwardly out of its lowest position, or if any of siderails **58**, **60** are moved out of their raised positions, or if the one or more casters **49** are released or unbraked, each of those unwanted conditions is considered to be a violation of the falls protocol by the bed operating software of control circuitry **110** assuming the falls protocol is enabled.

[0057] According to the pulmonary protocol example given herein, head section **80** of mattress support deck **74** of frame **40** is required to be elevated above a threshold angle as measured by angle sensor **118**. The threshold angle may be about 30 degrees, for example. In some embodiments, GUI **120** is used to adjust the threshold angle to an angle greater than 30 degrees (e.g., 45 degrees or 60 degrees) or to an angle less than 30 degrees (e.g., 17 degrees or 10 degrees). Thus, if the HOB angle falls below the threshold angle, that is considered to be a violation of the pulmonary protocol, assuming the pulmonary protocol is enabled. Having the HOB angle above a threshold angle, such as 30 degrees, helps to prevent ventilator assisted pneumonia (VAP) in some patients. Alternatively or additionally, the pulmonary protocol may require that the patient undergo certain pulmonary therapies performed by mattress **42** within certain time intervals. Examples of such pulmonary therapies include, for example, continuous lateral rotation therapy (CLRT) using rotation bladders of mattress **42** and/or percussion/vibration (P&V) therapy using P&V bladders of mattress **42**. Thus, CLRT or P&V therapy may be required for ½ hour, 1 hour, 2 hours, etc. twice per day or at least once within in any 8 hour nursing shift according to the pulmonary protocol as monitored by the bed operating software, just to give a couple of nonlimiting examples.

[0058] According to the safe skin protocol example given herein, the patient is required to move relative to the frame **40** by a threshold amount so as not to be stationary for a threshold amount of time. Such movement is detected by the PPM system **70** in the illustrative embodiment. Alternatively or additionally, the safe skin protocol may require the operation of certain features of mattress **42** for certain periods of time or at certain intervals. For example, an alternating pressure feature of mattress **42** may be required to be operating substantially continuously while the patient is in bed **12**. Alternating pressure refers to sequentially inflating and deflating every other laterally extending bladder of two sets of interleaved or interdigitated bladders of mattress **12**. Alternatively or additionally, the safe skin protocol may require a microclimate management layer or low airloss topper of mattress **42** to have air circulated therethrough to remove or reduce the amount of moisture and/or heat between the patient and the upper surface of mattress **42**. Further alternatively or additionally, the safe skin protocol may require that turn assist bladders of the mattress **42** be used from time to time to turn the patient to their left side and to their right side. When the safe skin protocol is enabled, the bed operating software of control circuitry **110** monitors the various time thresholds for patient movement, alternating pressure, microclimate management operation, and turn assist, as the case may be.

[0059] Referring now to FIG. **3**, beds **12** are shown diagrammatically as bed **12***a*, bed **12***b*, bed **12***c*, and bed **12***n*. Vertical dashes between bed **12***c* and bed **12***n* indicate diagrammatically that system **10** may include any number of beds **12** that may be upgraded with new bed operating software from upgrade server **22** or remote computer of upgrade system **23** via network **18**. Each bed **12***a*-**12***n* includes a memory toggle **140** that indicates whether memory bank **111** or memory bank **113** is the active memory bank **111**, **113** and which is the inactive memory bank **111**, **113**. In FIG. **3**, bed **12***b* is shown with upper and lower enlarged depictions of memory banks **111**, **113**. In the upper depiction of memory banks **111**, **113**, the memory toggle **140** is in a first position having memory bank **111** being the active memory bank and having memory bank **113** being the inactive memory bank. Thus, the bed operating software (Rev. 1.0 in the illustrative example) of memory bank **111** is used to operate the features and functions of bed **12***b* before the software upgrade as indicated by the textual information provided in connection with the upper pair of memory banks **111**, **113**. In the lower depiction of memory banks **111**, **113**, the memory toggle **140** is in a second position having memory bank **113** being the active memory bank and having memory bank **111** being the inactive memory bank. Thus, the bed operating software (Rev. 2.0 in the illustrative example) of memory bank **113** is used to operate the features and functions of bed **12***b* after the software upgrade as indicated by the textual information provided in connection with the lower pair of memory banks **111**, **113**.

[0060] Memory toggle **140** is a toggle flag stored in a particular memory location of memory **114** outside of memory banks **111**, **113** and is used at start-up (aka boot-up) to determine which of memory banks **111**, **113** is to be the active memory bank that contains the bed operating software to be used to operate the bed **12** after start-up. The use of memory toggle **140** allows new bed operating software to be loaded onto the inactive memory bank **111**, **113** while a prior version of bed operating software continues to be run on the active memory bank **111**, **113**. That is, bed **12** continues to operate according to the current version of bed operating software (Rev 1.0 in the illustrative example of FIG. **3**) while the new operating software (Rev. 2.0 in the illustrative example of FIG. **3**) is being loaded. This is an improvement over prior art systems, such as the one disclosed in U.S. Pat. No. 9,513,899, in which the bed **12** is rendered inoperable during the software upgrade process while the existing or current version of the operating software is being overwritten with the new version of the operating software.

[0061] Upon receiving the file transfer of the new bed operating software from the upgrade server **22** and/or computer **23**, the microprocessor **112** of the respective bed **12** changes the binary state of the memory toggle **140** (e.g., toggle flag) to change the respective memory bank **111**, **113** that is active. The memory toggle **140**, therefore, includes a single bit in some embodiments and includes multiple bits (such as a word) in other embodiments. Thus, if memory bank **111** was previously active prior to the software upgrade, then memory bank **113** becomes the active memory bank after the software upgrade at next start-up or boot-up of the respective bed **12**, and if memory bank **113** was previously active prior to the software upgrade, then memory bank **111** becomes the active memory bank after the software upgrade at the next start-up or boot-up of the respective bed **12**. However, the previous version of bed operating software remains in whichever memory bank **111**, **113** becomes the inactive memory bank **111**, **113** after the software upgrade.

[0062] According to the present disclosure, therefore, two versions of bed operating software are stored in memory **114** of bed **12** at any given time, assuming the bed operating software has been upgraded at least once. Because of this, if an error of sufficient severity occurs in the new version of bed operating software in the active memory bank **111**, **113**, the memory toggle **140** is switched so that the prior version of bed operating software is, once again, used to operate the features and functions of the bed. Such an error is sometimes referred to herein as a “bed operating software malfunction error” or just “malfunction error” or some such similar term. The switching of the memory toggle **140** in response to such a malfunction error occurs automatically at the respective bed **12** in some embodiments. In such embodiments, memory **114** includes software outside of memory banks **111**, **113** that is executed by processor **112** to determine whether to switch the memory toggle **140** to change the inactive memory bank **111**, **113** to the active memory bank **111**, **113** in response to the detected malfunction error.

[0063] In other embodiments, a user input such as one of inputs **106**, **108** or an input accessible using graphical display screen **120** is selected or pressed by a caregiver on the respective bed **12** to switch the memory toggle **140** to change the inactive memory bank **111**, **113** to the active memory bank **111**, **113** after detection of an operating software malfunction error of sufficient magnitude in the current bed operating software. In such embodiments, a message or alert regarding the malfunction error is displayed on screen **120** and/or is transmitted to a wireless communication device, such as tablet computer **38**, carried by an assigned caregiver and/or is displayed on a display screen of the upgrade server **22** and/or one remote upgrade computer **23** and/or on some other remote computer **23** such as a nurse call computer. Thus, in some embodiments, communication server **20** initiates a message to the caregiver's wireless communication device **38** in response to receiving a message originating from bed **12** that a bed operating software malfunction error has occurred in the active bed operating software.

[0064] Referring now to FIG. **4**, an illustrative example of an upgrade computer screen **150** which appears on the display screen of the upgrade server **22** and/or remote upgrade computer **23** and which is used by a user to select the beds **12***a*-**12***n* that are to be updated with new operating software is shown. Screen **150** includes a table **152** having a first column **154** that lists the bed identification codes (bed ID's) of the beds **12***a*-**12***n* that are in communication with server **22**, a second column **156** that lists the bed type of the beds **12***a*-**12***n* that are in communication with server **22**, a third column **158** that lists the location of each of beds **12***a*-**12***n* that are in communication with server **22**, and a fourth column **160** that lists the current version of software being operated in the active memory bank **111**, **113** of the respective beds, regardless of whether the first memory bank **111** or the second memory bank **113** is the active memory bank.

[0065] Screen **150** also has a separate column **162** to the right of table **152** that lists the version of software that is available for upgrading or updating each bed **12***a*-**12***n* in communication with server **22** and a separate Upgrade/Don't Upgrade column **164** to the right of column **162** that indicates, for each bed **12***a*-**12***n* in table **152**, whether the particular bed **12***a*-**12***n* is to have its software updated to the new version of software shown in column **162**. Each cell in column **164** can be selected by a user sequentially to toggle between the “Upgrade” and “Don't Upgrade” options. Thus, if a cell has the text “Upgrade” shown in it, then a subsequent selection of that cell will change the selection to “Don't Upgrade” and vice versa. If the display screen of the upgrade server **22** or computer **23** on which screen **150** is displayed is a touchscreen, then the user can simply touch each cell in column **164**, as desired, to toggle between the “Upgrade” and “Don't Upgrade” options. Alternatively or additionally, the user can click on each cell in column **164** with a cursor controlled by a computer mouse, for example, or use arrow keys to move a cursor to the desired cell and then pressing an enter key, to toggle between the two options. It should be appreciated that column **164** may just as well use the terms “Update” and “Don't Update” or similar such terms, in lieu of the terms “Upgrade” and “Don't Upgrade.”

[0066] In some embodiments, the list of beds appearing in table **152** includes those patient beds **12** that have registered with server **22** or remote computer **23**. For example, each of the patient beds **12** that register with the server **22** or computer **23** transmits a bed identification (ID) and a software version number of the current bed operating software for the respective patient bed. The registration occurs, in some embodiments, when a new patient bed **12** is initially communicatively coupled to respective bed connector **14**, for example. Server **22** communicates with server **24** in some embodiments to determine the location of the respective bed **12** if the bed **12** has an equipment tag (similar to badge **30** worn by the caregiver) that communicates with transceiver unit **28**.

[0067] Screen **150** has an Execute Upgrade button or icon **166** that is selected by the user after the desired Upgrade/Don't Upgrade selections have been made in column **164** for each listed bed **12***a*-**12***n* if the user wants to proceed with upgrading the bed operating software with the new version of software listed in column **162**. In response to selection of button **166**, server **22** transmits to each bed **12***a*-**12***n* having the “Upgrade” option selected in column **164**, the version of bed operating software listed in column **162** for that particular bed **12***a*-**12***n*. Thus, the beds **12***a*-**12***n* having the “Upgrade” option selected are upgraded with new bed operating software simultaneously. This is not to say that the software downloads start and end exactly at the same time for each bed **12***a*-**12***n*, just that the software download is initiated at the same time by selection of button **166**. Screen **150** has a Cancel icon or button **168** that is selected by the user if it is decided that none of the beds **12***a*-**12***n* listed in table **152** are to be upgraded with new operating software. After selection of Cancel icon **168**, screen **150** is replaced by some other screen, such as a Home screen or other administration screen (not shown) in some embodiments.

[0068] Considering the specific example shown in FIG. **4**, bed **12***a* and bed **12***b* which have the associated information shown in the first two rows of table **152**, each are currently operating using bed operating software version 1.0 and can be upgraded so as to operate according to software version 2.0 as indicated in the two upper cells of column **162**. Accordingly, the selections in column **164** for the upper two cells is “Upgrade” so that beds **12***a* and **12***b* will be upgraded to the version 2.0 bed operating software in response to selection of button **166** by the user. On the other hand, bed **12***c*, which is the same type of bed as indicated in column **156** as beds **12***a*, **12***b*, is already operating according to bed operating software version 2.0 and there is no new version of software to which bed **12***c* can be upgraded. Thus, the cell in column **162** corresponding to bed **12***c* has dashes, or is left blank or has some other null indicator in the cell, to indicate that there is no available software version for upgrading bed **12***c*. Accordingly, the cell in column **164** corresponding to bed **12***c* defaults to the “Don't Upgrade” option.

[0069] In some embodiments, subsequent selections of the cell in column **164** corresponding to bed **12***c* have no effect since there is no available software for bed **12***c* to be upgraded. That is, the “Don't Upgrade” option will persist in the cell despite subsequent selection of that particular cell by the user. As to bed **12***n*, table **152** indicates in column **156** that it is a “Type 2” bed and so bed **12***n* is not the same type (e.g., same make and/or model) as beds **12***a*, **12***b*, **12***c*. Furthermore, the current version of bed operating software for bed **12***n* is version 6.0 as indicated in column **160** of table **152** and the upgrade software version listed in column **162** for bed **12***n* is version 7.0. Thus, FIG. **4** suggests that bed **12***n* may be an older bed than the Type 1 beds **12***a*, **12***b*, **12***c* since its software version is a higher number version of software. In column **164**, the cell associated with bed **12***n* has the “Upgrade” option selected. Thus, when button **166** is selected, bed **12***n* will be upgraded with the version 7.0 bed operating software for Type 2 beds while beds **12***a*, **12***b* will be upgraded with version 2.0 bed operating software for Type 1 beds. Accordingly, it is contemplated by this disclosure that different types of beds are upgraded with respective different software versions simultaneously via selection of button **166**.

[0070] Referring now to FIG. **5**, an illustrative example of a remote toggle screen **170** which appears on the display screen of the upgrade server **22** and/or remote upgrade computer **23** and which is used by a user to toggle between the active and inactive memory banks **111**, **113** for beds experiencing operating software malfunction errors is shown. Screen **170** includes a table **172** having a first column **174** that lists the bed identification codes (bed ID's) of the beds **12***a*-**12***n* that are in communication with server **22** and that have experienced one or more bed operating software malfunction errors. In the illustrative example, beds **12***a* and **12***c* have experienced such errors. As alluded to above, the errors in the bed operating software that are considered to be malfunction errors are those of sufficient magnitude to affect the operation of the associated bed to such an extent that the version of operating software should be changed. Thus, in some embodiments, noncritical or minor errors that may occur during execution of the bed operating software may not be considered to be of sufficient magnitude to warrant a designation as a malfunction error.

[0071] Still referring to FIG. **5**, table **172** also includes a second column **176** that lists the bed type of the beds **12***a*-**12***n* that are in communication with server **22** and that have experienced malfunction errors and a third column **178** that lists the location of each of beds **12***a*-**12***n* that are in communication with server **22** and that have experienced malfunction errors. Table **172** also has a fourth column **180** and a fifth column **182** that lists the versions of bed operating software that are currently stored in the active and inactive memory banks **111**, **113**, respectively, of the respective beds in table **172**. In the illustrative example of FIG. **5**, bed **12***a* has memory bank **111** (i.e., the first memory bank or memory bank 1) as its active memory bank with version 3.0 being the bed operating software version in which one or more malfunction errors have occurred as indicated in column **180** and bed **12***a* has memory bank **113** (i.e., the second memory bank or memory bank 2) as its inactive memory bank with version 2.0 being the prior version of bed operating software stored therein as indicated in column **182**. Also in the illustrative example, bed **12***c* has memory bank **113** (i.e., the second memory bank or memory bank 2) as its active memory bank with version 4.0 being the bed operating software version in which one or more malfunction errors have occurred as indicated in column **180** and bed **12***c* has memory bank **111** (i.e., the first memory bank or memory bank 1) as its inactive memory bank with version 3.0 being the prior version of bed operating software stored therein as indicated in column **182**.

[0072] Screen **170** also has a separate Toggle/Don't Toggle column **184** to the right of table **172** that indicates, for each bed **12***a***-12***n* in table **172** (e.g., beds **12***a*, **12***c* in the illustrative example), whether the particular bed **12***a***-12***n* is to have its operating software version toggled to the version stored in the inactive memory bank listed in column **182**. Each cell in column **184** can be selected by a user sequentially to switch between the “Toggle” and “Don't Toggle” options. Thus, if a cell in column **184** has the text “Toggle” shown in it, then a subsequent selection of that cell will change the selection to “Don't Toggle” and vice versa. If the display screen of the upgrade server **22** or computer **23** on which screen **170** is displayed is a touchscreen, then the user can simply touch each cell in column **184**, as desired, to switch between the “Toggle” and “Don't Toggle” options. Alternatively or additionally, the user can click on each cell in column **184** with a cursor controlled by a computer mouse, for example, or use arrow keys to move a cursor to the desired cell and then pressing an enter key, to switch between the two options. It should be appreciated that column **184** may just as well use the terms other than “Toggle” and “Don't Toggle” but that have a similar meaning.

[0073] In the illustrative example, bed **12***c* has the “Don't Toggle” option selected because software version 3.0 in the inactive memory bank **111** of bed **12***c* is the same software version 3.0 in the active memory bank **111** of bed **12***a* which has experienced a malfunction error while operating in bed **12***a*. Thus, the user has decided not to toggle over to a software version for bed **12***c* that is a version which has proven in bed **12***a*, to already have a malfunction error.

[0074] Screen **170** has an Execute Toggle button or icon **186** that is selected by the user after the desired Toggle/Don't Toggle selections have been made in column **184** for each listed bed **12***a***-12***n* in table **174** if the user wants to proceed with toggling or switching the memory banks **111**, **113** between active and inactive states so that the bed operating software in the inactive memory bank **111**, **113**, as the case may be, becomes the active bed operating software. In response to selection of button **186**, server **22** transmits to each bed **12***a***-12***n* having the “Toggle” option selected in column **184**, a message or instruction to switch or toggle the memory banks **111**, **113** between the active and inactive states. Thus, the beds **12***a***-12***n* having the “Toggle” option selected are each toggled simultaneously. This is not to say that the toggling of memory banks **111**, **113** start and end exactly at the same time for each bed **12***a***-12***n*, just that the toggling is initiated at the same time by selection of button **186**. Screen **170** has a Cancel icon or button **188** that is selected by the user if it is decided that none of the beds **12***a***-12***n* listed in table **172** are to have their memory banks **111, 113** toggled or switched between the active and inactive states. After selection of Cancel icon **188**, screen **170** is replaced by some other screen, such as a Home screen or other administration screen (not shown) in some embodiments.

[0075] Although certain illustrative embodiments have been described in detail above, variations and modifications exist within the scope and spirit of this disclosure as described and as defined in the following claims.

Claims

1.-**10**. (canceled)
11. A patient bed for use with a network including a remote computer, the patient bed comprising a bed frame to support a patient, bed circuitry carried by the bed frame, the bed circuitry being configured to receive a new version of bed operating software from the remote computer, the bed circuitry including a first memory bank and a second memory bank, one of the first and second memory banks storing therein a current version of bed operating software, the other of the first and second memory banks storing therein the new version of bed operating software received from the remote computer, whereby two versions of bed operating software is stored in the bed circuitry, wherein the bed circuitry is programmed with a memory toggle that is used to select between the first and second memory banks thereby to select which version of bed operating software is used to operate the patient bed, a touchscreen display carried by the bed frame and coupled to the bed circuitry, and a user input viewable on the touchscreen display, the user input being touchable on the graphical display screen to manually switch the memory toggle so that a user is able to select which of the software versions stored in the first and second memory banks is used to operate the patient bed, wherein the memory toggle of the patient bed is also switchable in response to receiving a signal from the remote computer that is transmitted to the patient bed in response to an input at the remote computer, which input at the remote computer also results in signals being transmitted to other patient beds to switch respective memory toggles of the other patient beds, wherein the memory toggle comprises a toggle flag that is used at start-up of the patient bed to determine which of the first and second memory banks is currently active.
12. The patient bed of claim 11, wherein the first and second memory banks each comprise flash memory banks that are independent of each other.
13. The patient bed of claim 11, wherein the new version of bed operating software received from the remote computer is stored at download into whichever of the first and second memory banks is inactive.
14. The patient bed of claim 13, wherein upon next start-up of the patient bed, the memory toggle is switched so that a previously active memory bank of the first and second memory banks becomes inactive and so that a previously inactive memory bank of the first and second memory banks becomes active.
15. The patient bed of claim 13, wherein during download of the new version of bed operating software into the inactive memory bank, the patient bed continues to be operated according to the current version of bed operating software stored in the active memory bank.
16. The patient bed of claim 11, wherein after the patient bed is operated according to the new version of the bed operating software, the bed circuitry is programmed to automatically switch the memory toggle if an error occurs in the new version of bed operating software so that the prior version of bed operating software is once again used to operate the patient bed.
17. The patient bed of claim 11, wherein the remote computer is usable to download the new version of operating software to the patient bed while also downloading other new versions of operating software to the other patient beds.
18. The patient bed of claim 11, further comprising a barrier coupled to the frame and configured to block egress of the patient from the frame and wherein the touchscreen display is mounted to the barrier.
19. The patient bed of claim 18, wherein the barrier comprises a siderail.
20. A system comprising a computer, and a plurality of patient beds that are remote from the computer, each patient bed comprising a bed frame to support a patient, bed circuitry carried by the bed frame, and a touchscreen display carried by the bed frame and coupled to the bed circuitry, the bed circuitry of each bed being configured to receive a new version of bed operating software from the computer, the bed circuitry including a first memory bank and a second memory bank, one of the first and second memory banks storing therein a current version of bed operating software for

each bed, the other of the first and second memory banks storing therein the new version of bed operating software received from the remote computer, whereby two versions of bed operating software is stored in the bed circuitry of each bed, wherein the computer is configured to upgrade at least some of the plurality of patient beds with the new version of bed operating software simultaneously, wherein the computer is configured to permit a user to select which of the plurality of patient beds is to be upgraded with the new version of bed operating software and to deselect others of the patient beds that are not to be upgraded, wherein the bed circuitry of each patient bed is programmed with a memory toggle that is used to select between the first and second memory banks thereby to select which version of bed operating software is used to operate the respective patient bed, wherein respective user inputs are viewable on the touchscreen displays of each of the patient beds, the respective user inputs being touchable on the corresponding touchscreen display to manually switch the memory toggle of the respective patient bed so that the user is able to select which of the software versions stored in the first and second memory banks of the respective bed is used to operate the respective patient bed, wherein the computer is also configured to receive an input from the user to remotely and simultaneously send signals to switch the memory toggles of the plurality of patient beds, wherein the memory toggle of each patient bed comprises a toggle flag that is used at start-up of the respective patient bed to determine which of the first and second memory banks is currently active.
21. The system of claim 20, wherein the computer is configured to display a list of each of the plurality of patient beds that is available for upgrade.
22. The system of claim 20, wherein each of the patient beds that are available for upgrade register with the computer.
23. The system of claim 22, wherein each of the patient beds that register with the computer transmits a bed identification (ID) and a software version number of the current bed operating software for the respective patient bed.
24. The system of claim 20, wherein the computer displays a table in which the user designates which of the plurality of patient beds is to be upgraded with the new version of bed operating software and to deselect the others of the patient beds that are not to be upgraded.
25. The system of claim 20, wherein the first and second memory banks of each patient bed each comprise flash memory banks that are independent of each other.
26. The system of claim 20, wherein the new version of bed operating software received from the remote computer by each patient bed is stored at download into whichever of the first and second memory banks is inactive in the respective patient bed.
27. The system of claim 26, wherein upon next start-up of each of the patient beds, the memory toggle is switched so that a previously active memory bank of the first and second memory banks becomes inactive and so that a previously inactive memory bank of the first and second memory banks becomes active.
28. The system of claim 26, wherein during download of the new version of bed operating software into the inactive memory bank of each patient bed, each of the patient beds continue to be operated according to the current version of bed operating software stored in the active memory bank of the respective patient bed.
29. The system of claim 20, wherein after each of the patient beds is operated according to the new version of the bed operating software, the respective bed circuitry is programmed to automatically switch the memory toggle if an error occurs in the new version of bed operating software so that the prior version of bed operating software is once again used to operate the respective patient bed.
30. The system of claim 20, wherein each patient bed further comprises a barrier coupled to the respective bed frame and configured to block egress of the corresponding patient from the respective bed frame and wherein each touchscreen display is mounted to the respective barrier.
31. The system of claim 30, wherein each barrier comprises a siderail.
32. A system comprising a computer, and a plurality of patient beds that are remote from the

computer, each patient bed comprising a bed frame to support a patient, bed circuitry carried by the bed frame, and a touchscreen display carried by the bed frame and coupled to the bed circuitry, the bed circuitry of each bed being configured to receive a new version of bed operating software from the computer, the bed circuitry including a first memory bank and a second memory bank, one of the first and second memory banks storing therein a current version of bed operating software for each bed, the other of the first and second memory banks storing therein the new version of bed operating software received from the remote computer, whereby two versions of bed operating software is stored in the bed circuitry of each bed, wherein the computer is configured to upgrade at least some of the plurality of patient beds with the new version of bed operating software simultaneously, wherein the computer is configured to permit a user to select which of the plurality of patient beds is to be upgraded with the new version of bed operating software and to deselect others of the patient beds that are not to be upgraded, wherein the bed circuitry of each patient bed is programmed with a memory toggle that is used to select between the first and second memory banks thereby to select which version of bed operating software is used to operate the respective patient bed, wherein respective user inputs are viewable on the touchscreen displays of each of the patient beds, the respective user inputs being touchable on the corresponding touchscreen display to manually switch the memory toggle of the respective patient bed so that the user is able to select which of the software versions stored in the first and second memory banks of the respective bed is used to operate the respective patient bed, wherein the computer is also configured to receive an input from the user to remotely and simultaneously send signals to switch the memory toggles of the plurality of patient beds, wherein the remote computer is configured to display which of the first and second memory banks of the plurality of patient beds is active and which is inactive.